Real Estate Price Forecasting

Machine Learning for Property Valuation & Investment Strategy

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Agenda

1. Business Context
2. Dataset Overview
3. Exploratory Data Analysis
4. Statistical Analysis
5. Model Development & Results
6. Investment Strategy
7. Conclusions & Future Work

Why Automated Valuation Matters

The Challenge

- Traditional appraisals take **2–4 weeks** and cost $300–500
- Human bias introduces **up to 10% variance** between appraisers
- Market conditions change faster than manual processes

Target Metrics

KPI	Target
R^2 Score	> 0.90
MAPE	$< 12\%$
Latency	< 200ms

Our Solution

A machine learning model that provides **instant, objective valuations** with less than **11% average error**, enabling faster decisions for buyers, sellers, and lenders.

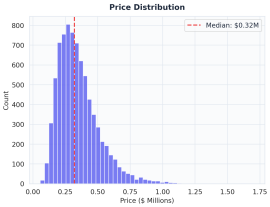
Dataset at a Glance

Key Statistics

- **8,000** real estate transactions
- **20** features (property, location, temporal)
- Sales from **2019 Q1 to 2025 Q4**
- **8 neighborhoods** across metro area
- Price range: $57K – $1.7M
- Median price: $320,800

Feature Groups

Property	SqFt, Bedrooms, Year Built, Type
Location	Neighborhood, Metro dist., Crime



Price by Neighborhood

Key Findings

- **Waterfront** commands the highest premium (1.5× base)
- **Downtown & Tech Hub** follow as high-value areas
- **Suburbs & Old Town** are the most affordable
- ANOVA: $p < 0.001$ confirms significant price differences

Price vs. Square Footage

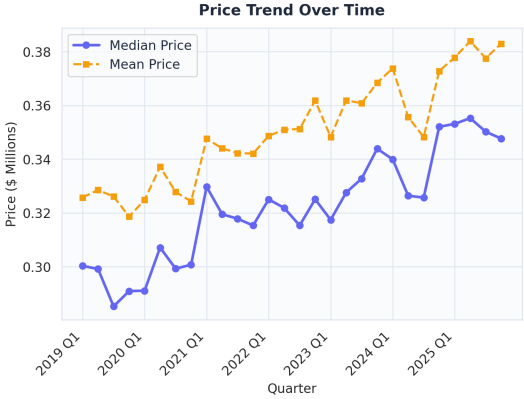
Analysis

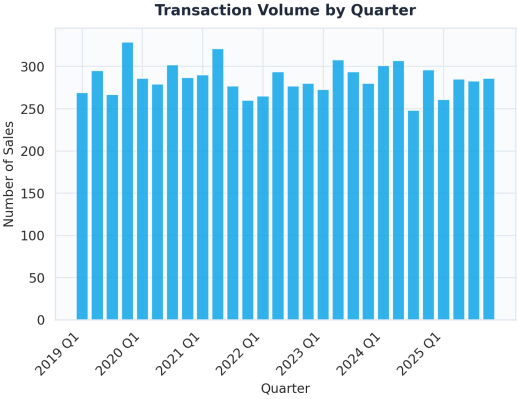
- Strong positive correlation ($r = 0.78$)
- Non-linear relationship — polynomial fit captures the curve
- Color gradient shows bedroom count adds incremental value

Insight

Each additional 100 sqft adds approximately $15K–25K in value, with diminishing returns above 4,000 sqft.

Price Trends & Transaction Volume





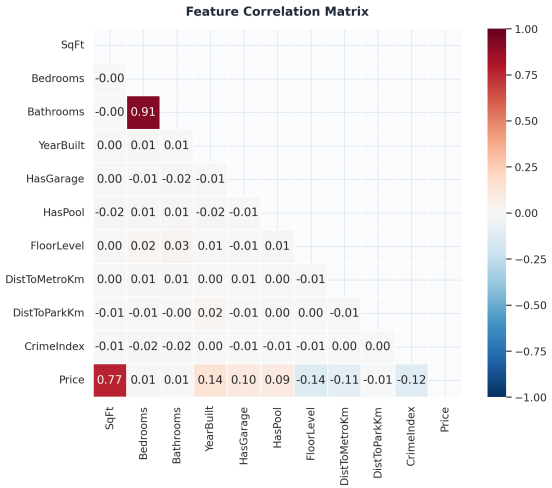
Price Appreciation

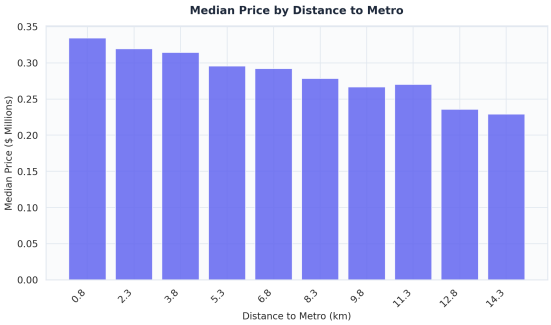
Steady upward trend with **3–4% annual appreciation** across the metro area.

Volume

Relatively stable transaction volume indicates a healthy, liquid market.

Correlation Matrix & Metro Proximity





Metro Premium

Hypothesis Testing Summary

Test	Statistic	p-value	Effect
ANOVA: Price $\sim$ Neighborhood	F-test	$p < 0.001$	Large (η^2)
Pearson: SqFt vs. Price	$r = 0.78$	$p < 0.001$	Strong
Spearman: Metro Dist. vs. Price	$\rho = -0.15$	$p < 0.001$	Moderate
Welch's t: Garage premium	t-test	$p < 0.001$	Significant
Kruskal-Wallis: Property Type	H-test	$p < 0.001$	Significant

Conclusion

All hypothesized price drivers show **statistically significant** relationships. Square footage and neighborhood are the dominant factors, followed by proximity to transit and property amenities.

Modeling Approach

8,000 Transactions → **Feature Engineering** → **Log-Price Transform** → **5 Models Trained** → **5-Fold Evalua**

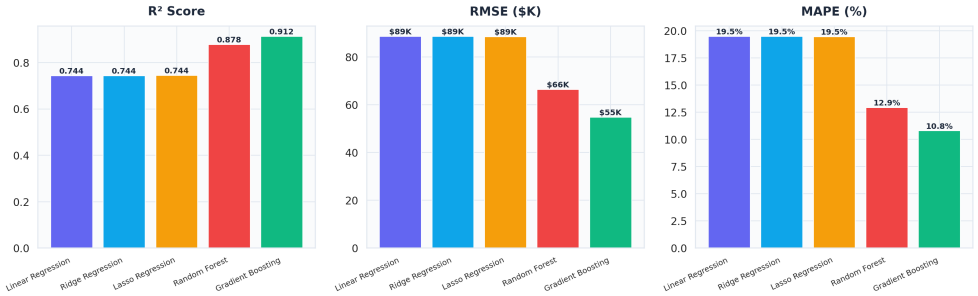
Engineered Features

- `Age` — property age in years
- `RoomDensity` — rooms per sqft
- `TotalAmenities` — amenity count
- `MetroAccess` — inverse metro dist.

Models Compared

1. Linear Regression
2. Ridge Regression ($\alpha = 1.0$)
3. Lasso Regression ($\alpha = 0.001$)
4. Random Forest (200 trees)
5. **Gradient Boosting** (200 trees)

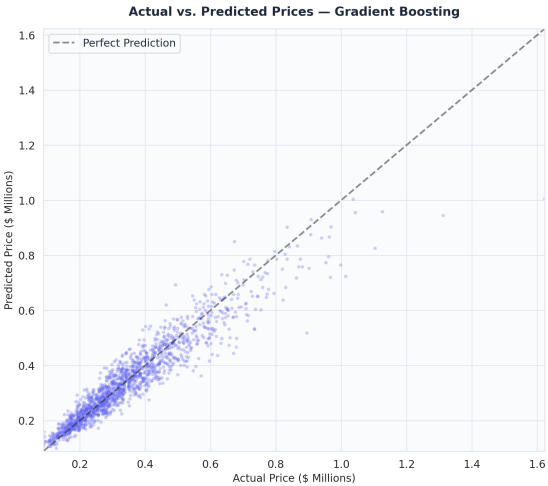
Model Comparison

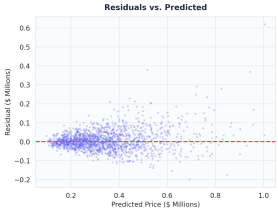


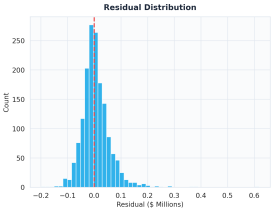
Winner: Gradient Boosting

$R^2 = 0.913$ RMSE = \$54,724 MAE = \$35,281 MAPE = 10.8%

Actual vs. Predicted & Residuals

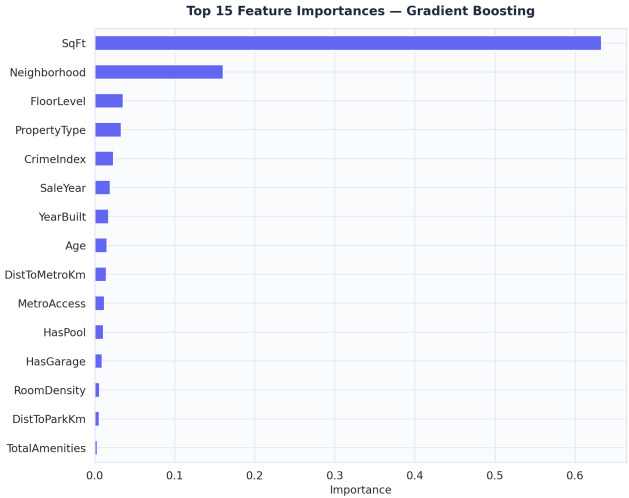






Diagnostic Summary

Feature Importance



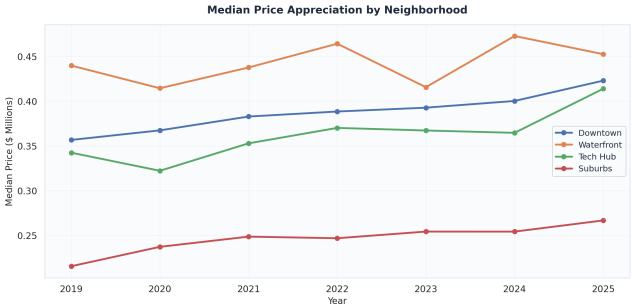
Top Price Drivers

1. **Square Footage** — dominant predictor
2. **Neighborhood** — location premium
3. **Metro Access** — transit proximity
4. **Year Built** — recency premium
5. **Total Amenities** — garage, pool, elevator

Takeaway

Physical size and location account for the majority of price variance —

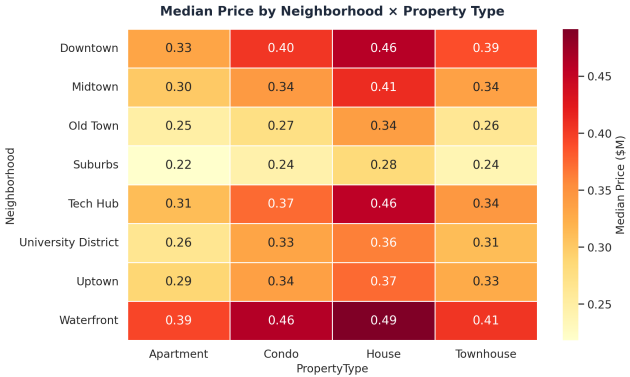
Neighborhood Appreciation Trends



Growth Leaders

Waterfront	Highest absolute appreciation, limited supply
Tech Hub	Fastest relative growth, driven by employment
Downtown	Steady premium, mature market
Suburbs	Affordable entry, remote-work tailwind

Investment Matrix



Strategy Recommendations

Growth Tech Hub condos & apartments — ride the appreciation wave

Value Suburb houses — best sqft/dollar with upside

Premium Waterfront houses — trophy assets with stability

Yield University District apartments — high rental demand

Risk Factors

Summary of Results

Achievements

- Gradient Boosting: $\mathbf{R^2 = 0.913}$
- Mean error: **$35K** (10.8% MAPE)
- All targets met ($R^2 > 0.90$, MAPE $< 12\%$)
- Identified top 5 price drivers
- Produced actionable investment insights

Future Roadmap

1. **Q2 '26:** SHAP explainability dashboard
2. **Q3 '26:** Geospatial kriging models
3. **Q3 '26:** Image-based CNN valuation
4. **Q4 '26:** FastAPI deployment
5. **Q1 '27:** Time series forecasting with LSTM

Thank You

Questions & Discussion

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